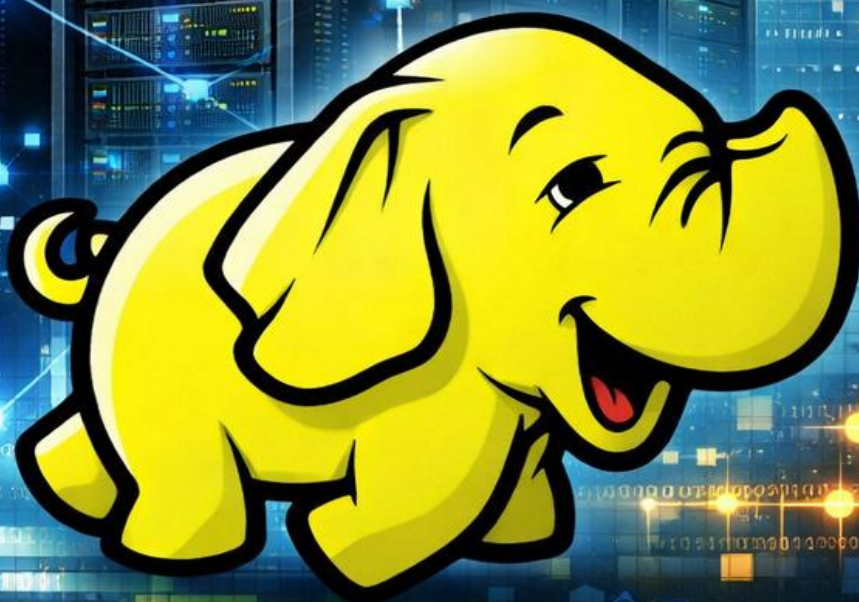




DESPLIEGUE DE UN CLÚSTER HADOOP MULTINODO



Ivana Sánchèz Pérez

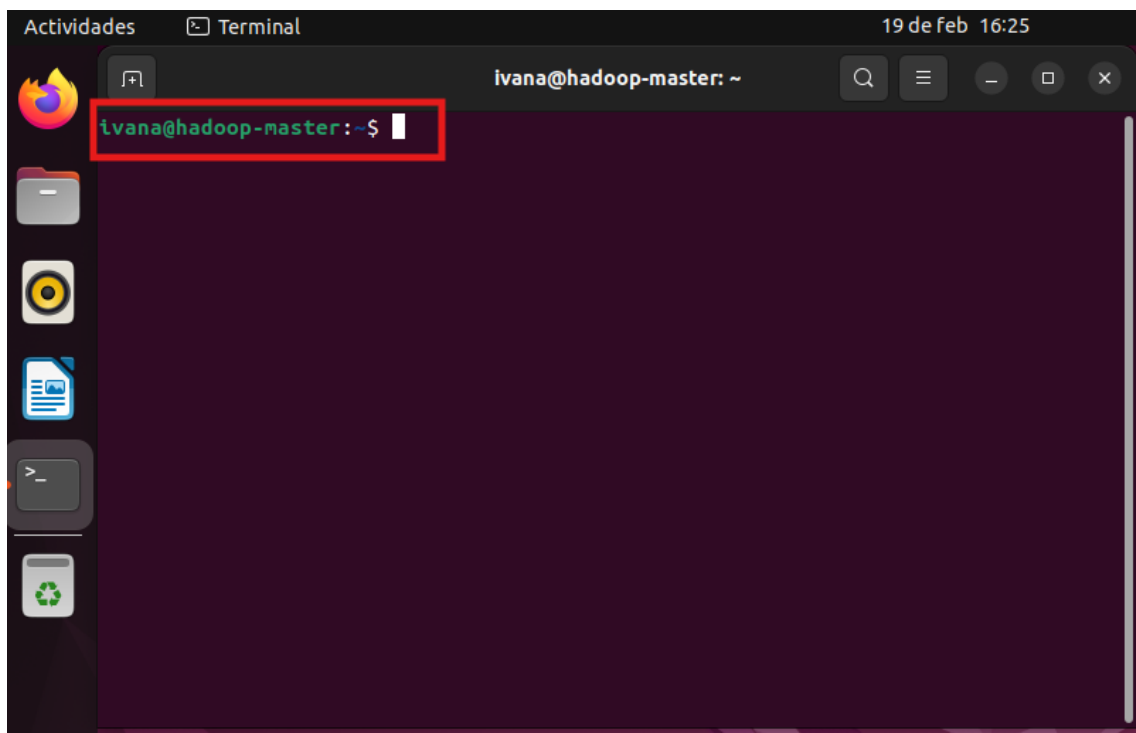
DESPLIEGUE DE UN CLÚSTER HADOOP MULTINODO

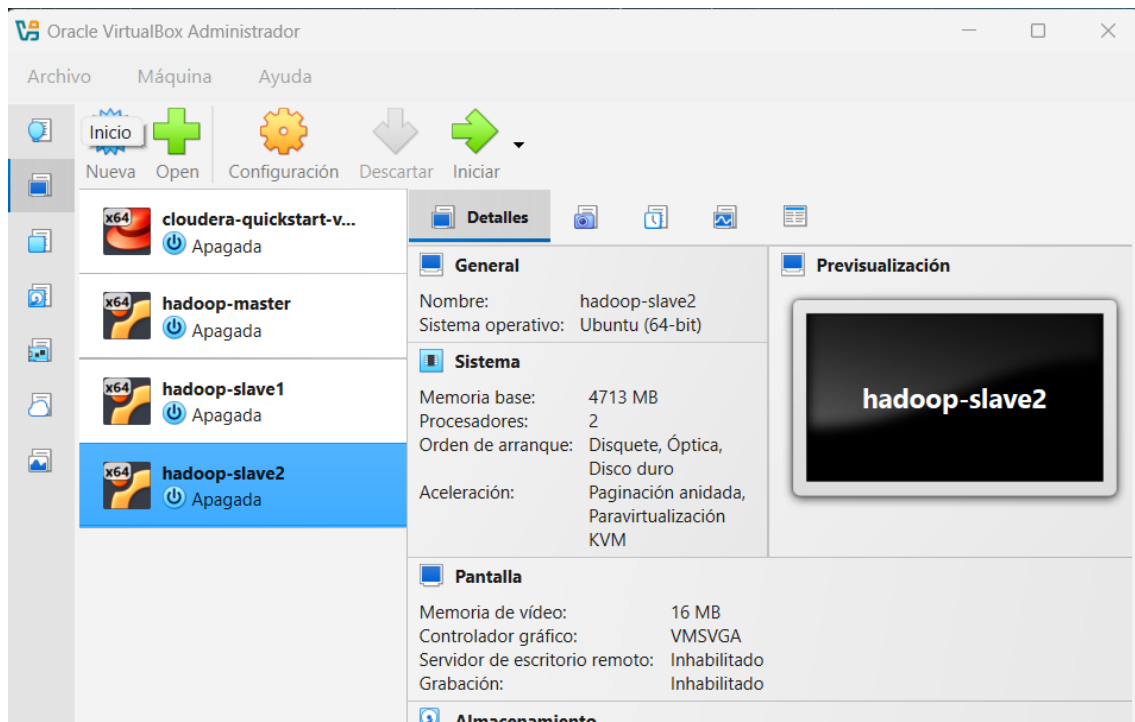
1. INTRODUCCIÓN

El objetivo de esta práctica es evolucionar una configuración de Hadoop de modo “Pseudo-clúster” a un entorno “Multinodo” real. Para ello, se han configurado tres máquinas virtuales: un nodo maestro (hadoop-master) y dos nodos esclavos (hadoop-slave1 y hadoop-slave2).

2. PREPARACIÓN DE LAS MÁQUINAS VIRTUALES

- a) Se cambia el nombre de la maquina principal y se crean dos clonaciones que actuarán como nodos esclavos.

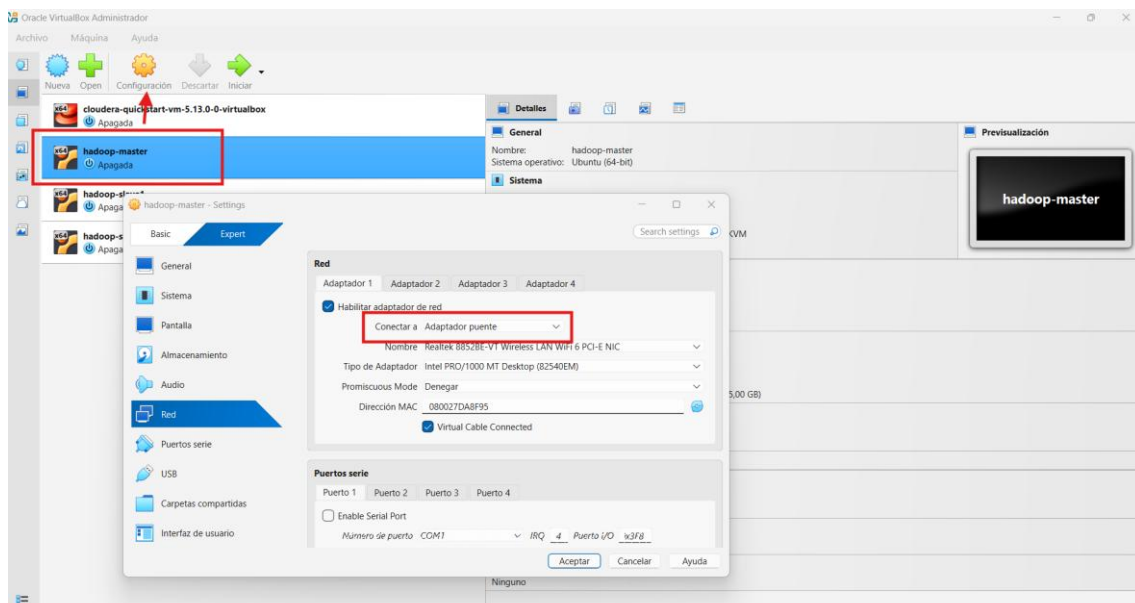




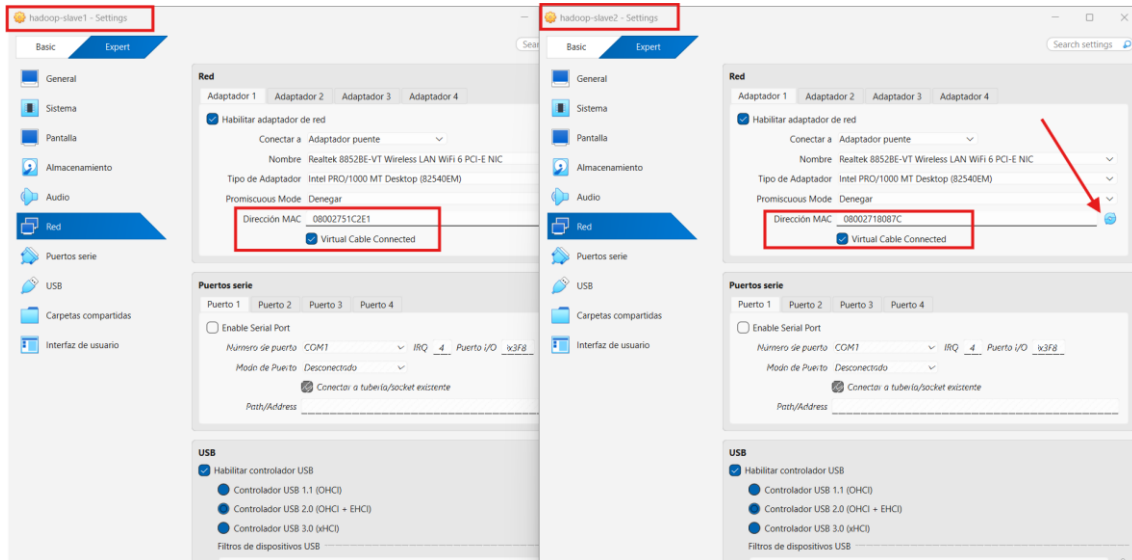
- b) Al clonar las máquinas, todas heredan la misma configuración de red, incluido el identificador que solicita la IP al router (o servidor DHCP).
- c) Para evitar conflictos, se generan nuevas direcciones MAC en las máquinas esclavas y se configura la red para que las tres máquinas puedan comunicarse entre sí

Con las máquinas apagadas, en cada una se realiza lo siguiente:

- **Adaptador de Red** → En la configuración de red de VirtualBox se selecciona el modo “Adaptador puente”: **Configuración > Red > Adaptador puente.**



- **Direcciones MAC** → Se generan nuevas direcciones MAC para las máquinas esclavas

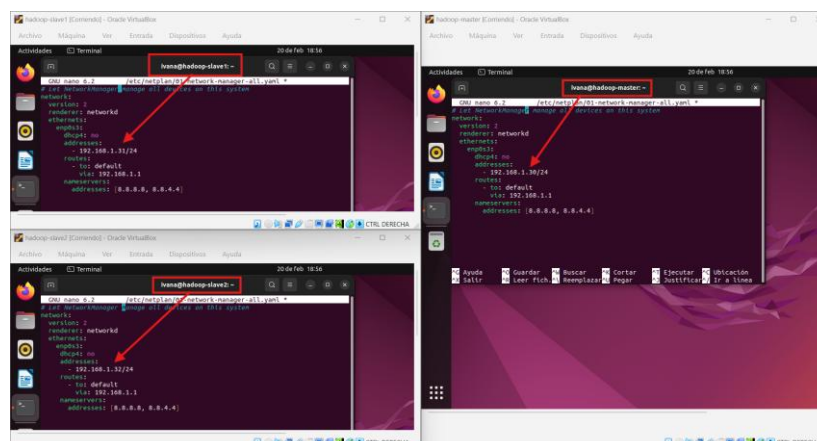


IP estáticas → Dentro de cada máquina se configuran las IP manualmente.

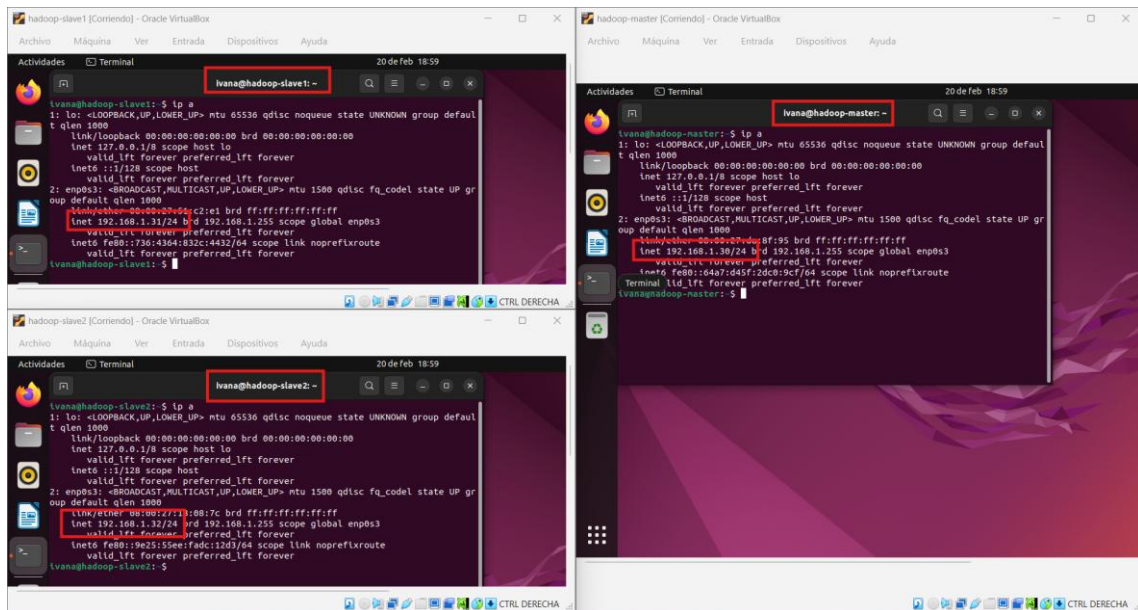
a) Primero se identifica la interfaz de red activa: ip a

```
lvana@hadoop-master: ~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:da18:f95 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.20/24 brd 192.168.0.255 scope global dynamic noprefroute enp0s3
        valid_lft 86345sec preferred_lft 86345sec
    inet fe80::6487:d45f:2dc0:9cf04 scope link noprefroute
        valid_lft forever preferred_lft forever
lvana@hadoop-master: ~$
```

b) A continuación, editamos el fichero **/etc/netplan/01-network-manager-all.yaml** para asignar las IP estáticas correspondientes en cada máquina.



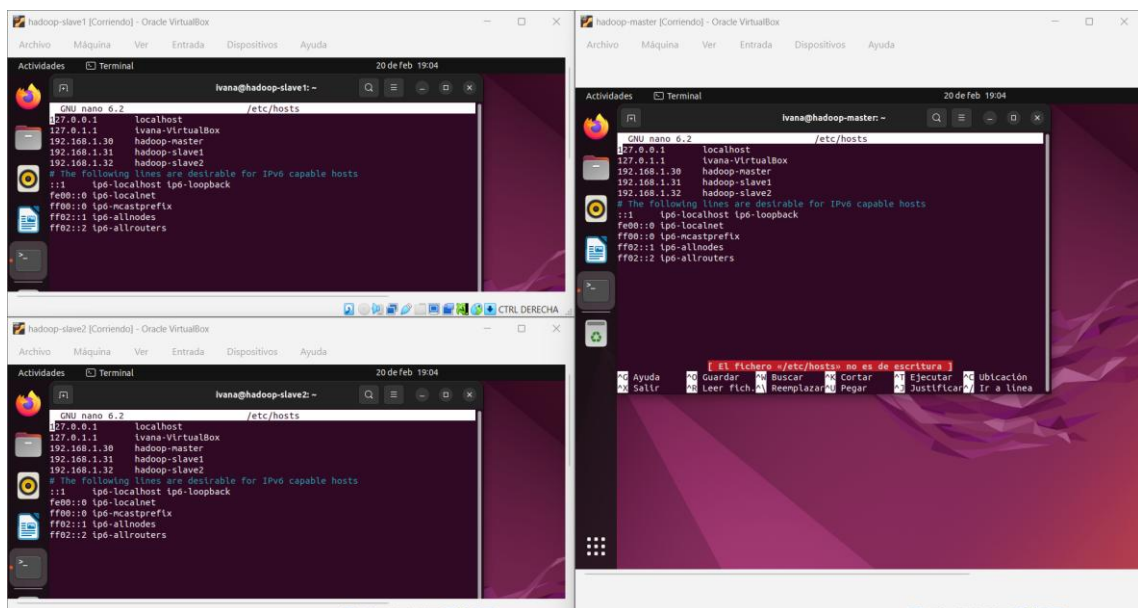
- c) Guardamos y aplicamos los cambios con: **sudo netplan apply**. Volvemos a hacer un **ip a** para comprobar que las IP son correctas.



3. CONFIGURACIÓN DE HOSTS Y ACCESO SSH

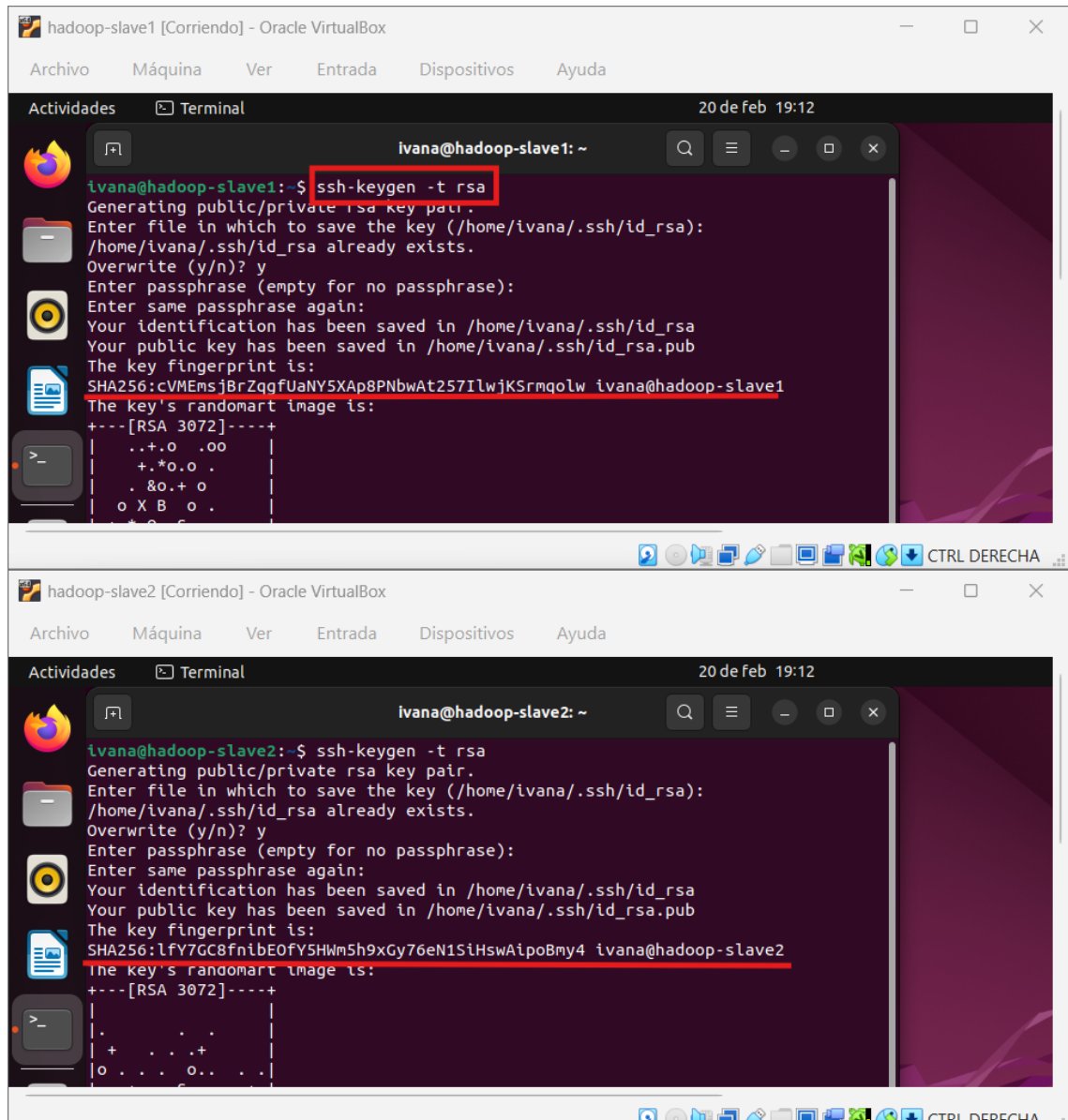
3.1. Archivo /etc/hosts

En cada máquina se añaden todas las IP del clúster, asociadas a los nombres hadoop-master, hadoop-slave1 y hadoop-slave2, en el fichero **/etc/hosts**.



3.2. Configuración de SSH sin contraseña

En las máquinas esclavas se genera una nueva clave SSH con **ssh-keygen -t rsa**, sobrescribiendo la anterior para evitar conflictos.



```
hadoop-slave1 [Corriendo] - Oracle VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

Actividades  Terminal  20 de feb 19:12

ivana@hadoop-slave1: ~
ivana@hadoop-slave1:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ivana/.ssh/id_rsa):
/home/ivana/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ivana/.ssh/id_rsa
Your public key has been saved in /home/ivana/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:cVMEMsjBrZqgfUaNY5XAp8PNbwAt257IlwJKSrmqolw ivana@hadoop-slave1
The key's randomart image is:
+---[RSA 3072]-----+
| ..+.o .oo |
| +.*o.o . |
| .&o.+ o |
| o X B o . |
| ..+.o .oo |
+---[RSA 3072]-----+

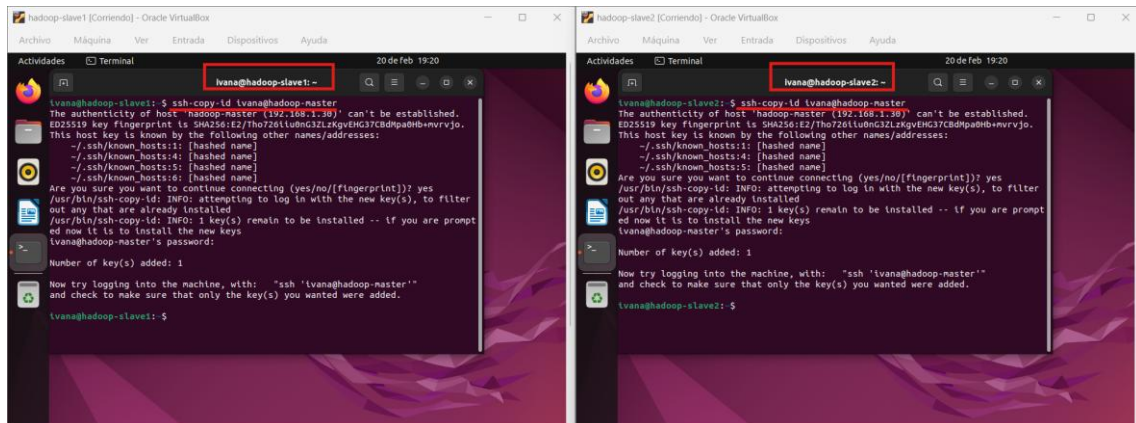
hadoop-slave2 [Corriendo] - Oracle VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

Actividades  Terminal  20 de feb 19:12

ivana@hadoop-slave2: ~
ivana@hadoop-slave2:~$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ivana/.ssh/id_rsa):
/home/ivana/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ivana/.ssh/id_rsa
Your public key has been saved in /home/ivana/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:lfY7GC8fnibE0fY5HwM5h9xGy76eN1SiHswAipoBmy4 ivana@hadoop-slave2
The key's randomart image is:
+---[RSA 3072]-----+
| . . . . . |
| + . . . + |
| o . . . o . . |
+---[RSA 3072]-----+
```

Después se configura el acceso sin contraseña entre al maestro y los esclavos así, autorizamos a la MV Master para que pueda entrar en las VMs esclavas y viceversa.

- Desde cada esclavo se copia la clave pública al maestro con:
ssh-copy-id ivana@hadoop-master

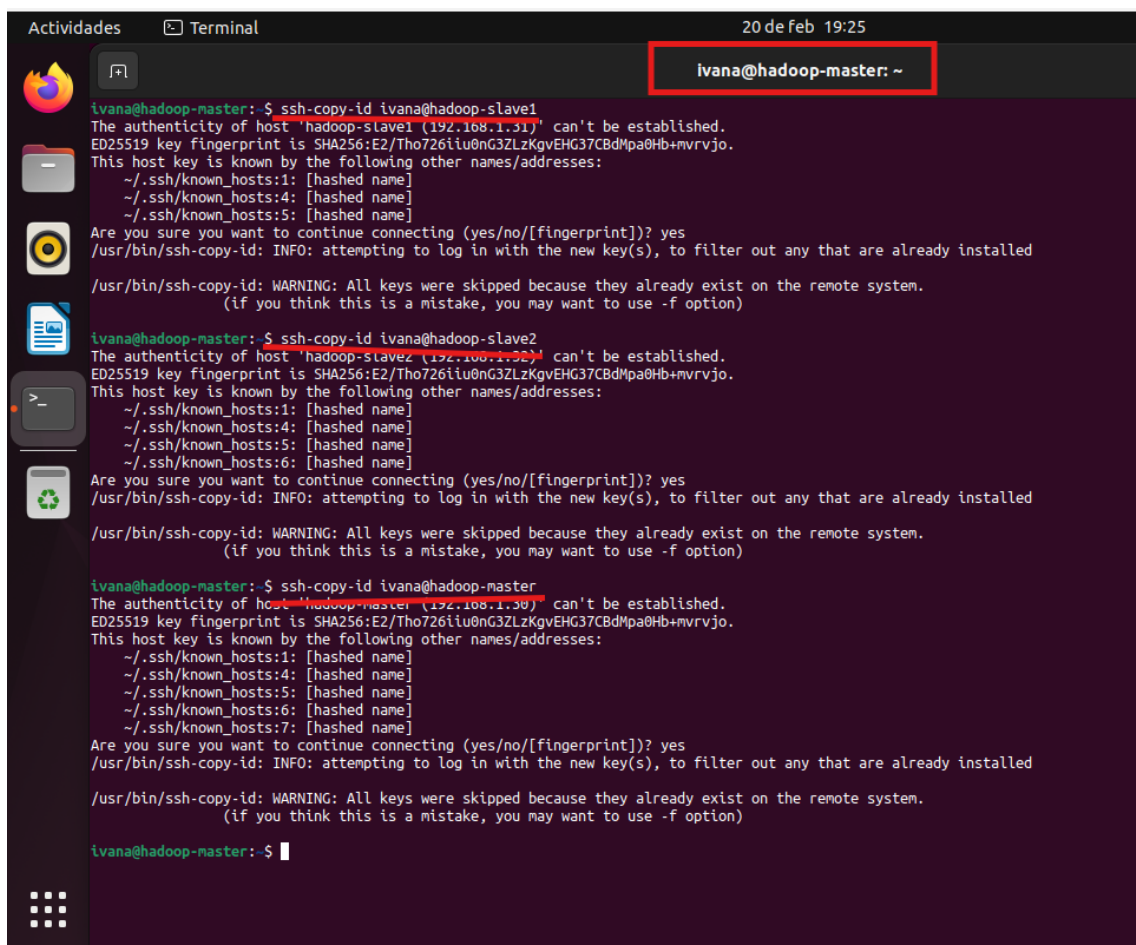


- Desde el Maestro se copia su clave a cada esclavo y también a sí mismo:

ssh-copy-id ivana@hadoop-slave1

ssh-copy-id ivana@hadoop-slave2

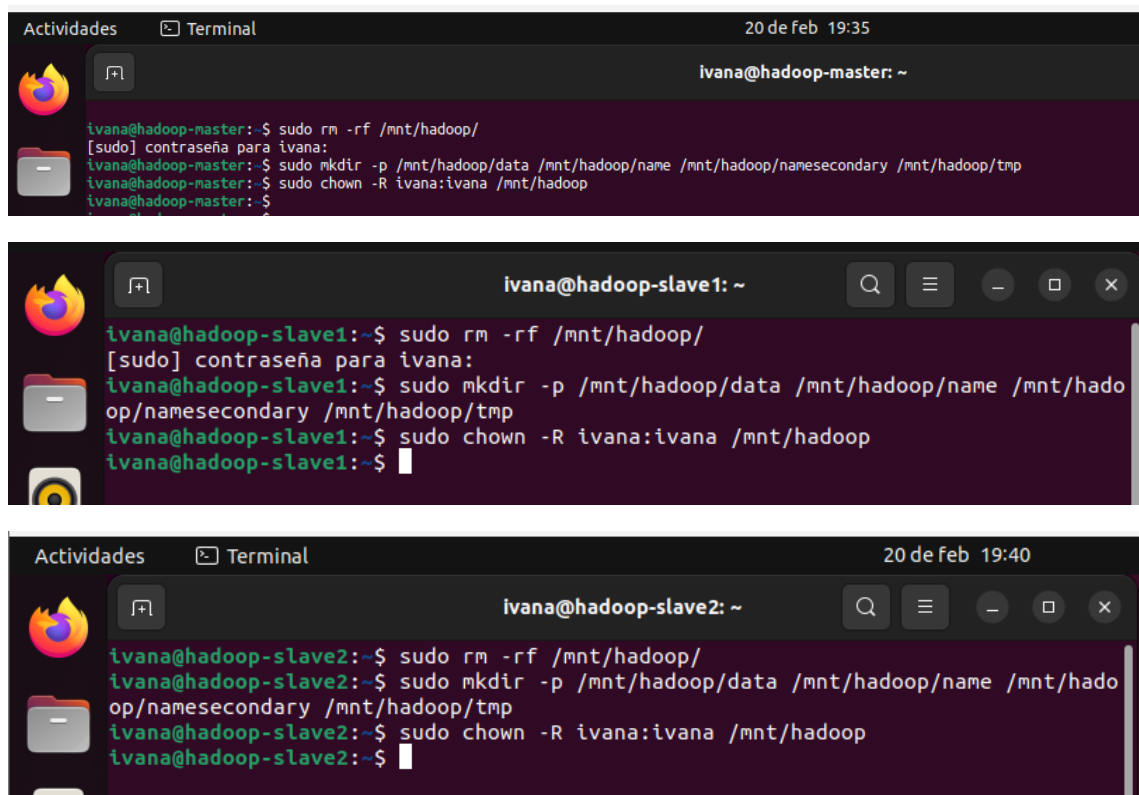
ssh-copy-id ivana@hadoop-master (a sí mismo).



4. LIMPIEZA DE DIRECTORIOS Y PREPARACIÓN DE HDFS

Como las máquinas son clones de un pseudo-clúster anterior, contienen datos antiguos que pueden provocar conflictos de ID en el nuevo entorno multimodo. Para evitarlo, en cada máquina se ejecutarán los siguientes comandos:

- 1- **`sudo rm -rf /mnt/hadoop/`**.
- 2- **`sudo mkdir -p /mnt/hadoop/data /mnt/hadoop/name /mnt/hadoop/namesecondary /mnt/hadoop/tmp`**.
- 3- **`sudo chown -R ivana:ivana /mnt/Hadoop`**



The image displays three terminal windows showing the execution of commands to clean and prepare the HDFS environment on different nodes.

Terminal 1 (Master Node): The terminal shows the user `ivana@hadoop-master` at 19:35. The commands executed are:

```
ivana@hadoop-master:~$ sudo rm -rf /mnt/hadoop/
[sudo] contraseña para ivana:
ivana@hadoop-master:~$ sudo mkdir -p /mnt/hadoop/data /mnt/hadoop/name /mnt/hadoop/namesecondary /mnt/hadoop/tmp
ivana@hadoop-master:~$ sudo chown -R ivana:ivana /mnt/hadoop
ivana@hadoop-master:~$
```

Terminal 2 (Slave Node 1): The terminal shows the user `ivana@hadoop-slave1` at 19:35. The commands executed are:

```
ivana@hadoop-slave1:~$ sudo rm -rf /mnt/hadoop/
[sudo] contraseña para ivana:
ivana@hadoop-slave1:~$ sudo mkdir -p /mnt/hadoop/data /mnt/hadoop/name /mnt/hadoop/namesecondary /mnt/hadoop/tmp
ivana@hadoop-slave1:~$ sudo chown -R ivana:ivana /mnt/hadoop
ivana@hadoop-slave1:~$
```

Terminal 3 (Slave Node 2): The terminal shows the user `ivana@hadoop-slave2` at 19:40. The commands executed are:

```
ivana@hadoop-slave2:~$ sudo rm -rf /mnt/hadoop/
ivana@hadoop-slave2:~$ sudo mkdir -p /mnt/hadoop/data /mnt/hadoop/name /mnt/hadoop/namesecondary /mnt/hadoop/tmp
ivana@hadoop-slave2:~$ sudo chown -R ivana:ivana /mnt/hadoop
ivana@hadoop-slave2:~$
```

Con esto se deja preparada la estructura de directorios limpia para el nuevo clúster.

5. CONFIGURACIÓN MULTINODO EN EL NODO MAESTRO

A continuación, se configuran los ficheros de Hadoop en el nodo maestro para que se utilice el nombre del clúster en lugar de localhost. Primero se accede al directorio de configuración:

`cd /usr/share/hadoop/etc/hadoop/`

Desde dicha carpeta editamos y modificamos los principales archivos:

1. **core-site.xml:** Cambia `hdfs://localhost:9000` por `hdfs://hadoop-master:9000`.



The screenshot shows a terminal window titled "Terminal" with the date and time "20 de feb 19:45". The user is logged in as "ivana@hadoop-master" and is in the directory "/usr/share/hadoop/etc/hadoop". The terminal shows the GNU nano 6.2 editor editing the file "core-site.xml". The content of the file is as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

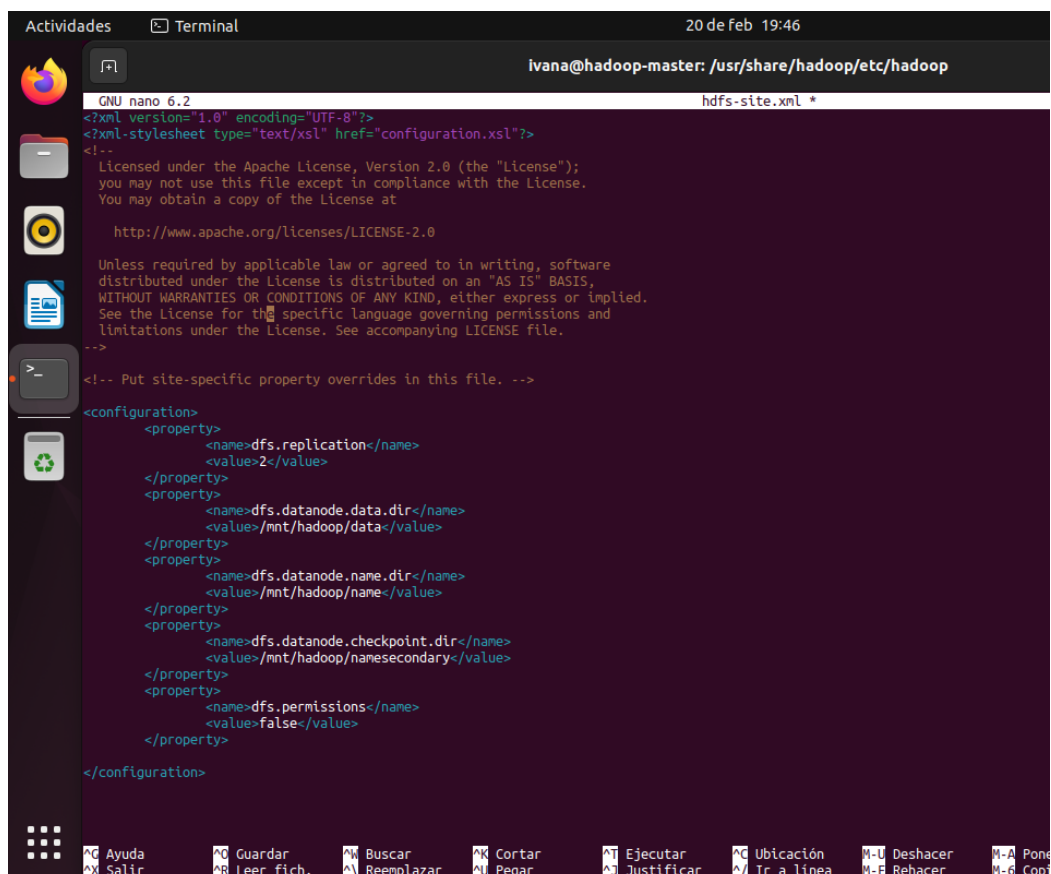
    http://www.apache.org/licenses/LICENSE-2.0

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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://hadoop-master:9000</value>
  </property>
  <property>
    <name>hadoop.tmp.dir</name>
    <value>/mnt/hadoop/tmp</value>
  </property>
</configuration>
```

2. **hdfs-site.xml:** Se cambia el valor de `dfs.replication` de 1 a 2 para que los datos se repliquen en dos nodos.



The screenshot shows a terminal window titled "Terminal" with the date and time "20 de feb 19:46". The user is logged in as "ivana@hadoop-master" and is in the directory "/usr/share/hadoop/etc/hadoop". The terminal shows the GNU nano 6.2 editor editing the file "hdfs-site.xml". The content of the file is as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="configuration.xsl"?>
<!--
  Licensed under the Apache License, Version 2.0 (the "License");
  you may not use this file except in compliance with the License.
  You may obtain a copy of the License at

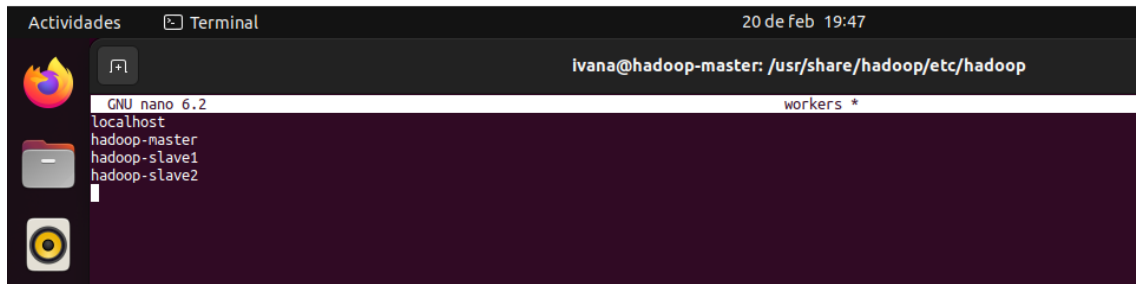
    http://www.apache.org/licenses/LICENSE-2.0

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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License. See accompanying LICENSE file.
-->

<!-- Put site-specific property overrides in this file. -->

<configuration>
  <property>
    <name>dfs.replication</name>
    <value>2</value>
  </property>
  <property>
    <name>dfs.datanode.data.dir</name>
    <value>/mnt/hadoop/data</value>
  </property>
  <property>
    <name>dfs.datanode.name.dir</name>
    <value>/mnt/hadoop/name</value>
  </property>
  <property>
    <name>dfs.datanode.checkpoint.dir</name>
    <value>/mnt/hadoop/name/secondary</value>
  </property>
  <property>
    <name>dfs.permissions</name>
    <value>false</value>
  </property>
</configuration>
```

3. **workers:** Edita el archivo workers para agregar los nombres de los esclavos (y el master si quieres que también almacene los datos)



A screenshot of a terminal window titled 'Actividades' and 'Terminal'. The terminal shows the user 'ivana' at 'hadoop-master' in the directory '/usr/share/hadoop/etc/hadoop'. The file 'workers' is being edited with 'GNU nano 6.2'. The content of the file is as follows:

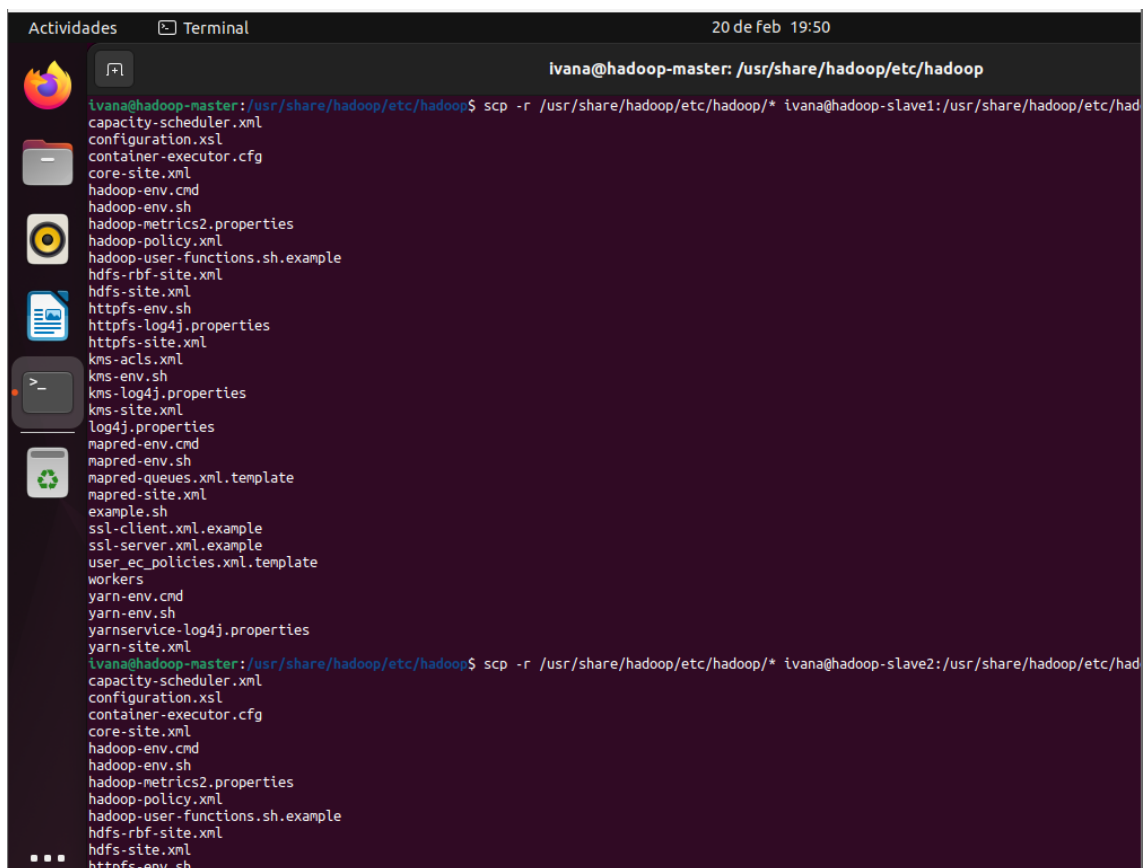
```
localhost
hadoop-master
hadoop-slave1
hadoop-slave2
```

6. SINCRONIZACIÓN DE CONFIGURACIÓN Y FORMATO DEL NAMENODE

Una vez configurado el maestro, se copia la configuración a los nodos esclavos:

```
scp -r /usr/share/hadoop/etc/hadoop/* ivana@hadoop-slave1:/usr/share/hadoop/etc/hadoop/
```

```
scp -r /usr/share/hadoop/etc/hadoop/* ivana@hadoop-slave2:/usr/share/hadoop/etc/hadoop/
```



A screenshot of a terminal window titled 'Actividades' and 'Terminal'. The terminal shows the user 'ivana' at 'hadoop-master' in the directory '/usr/share/hadoop/etc/hadoop'. The user has executed two scp commands to copy the Hadoop configuration files to the slave nodes. The first command is:

```
ivana@hadoop-master:/usr/share/hadoop/etc/hadoop$ scp -r /usr/share/hadoop/etc/hadoop/* ivana@hadoop-slave1:/usr/share/hadoop/etc/hadoop/
```

The second command is:

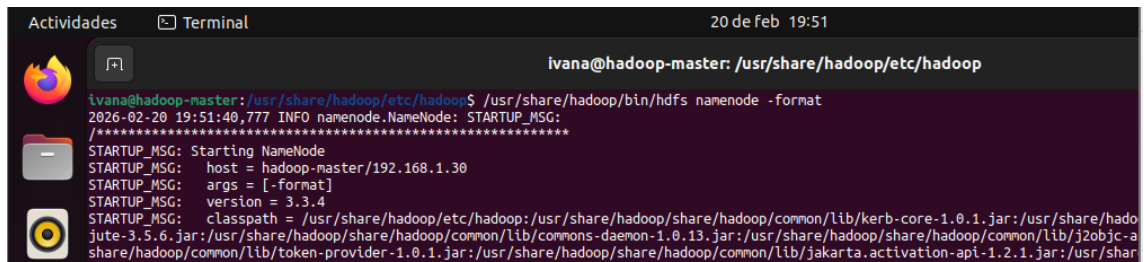
```
ivana@hadoop-master:/usr/share/hadoop/etc/hadoop$ scp -r /usr/share/hadoop/etc/hadoop/* ivana@hadoop-slave2:/usr/share/hadoop/etc/hadoop/
```

The terminal output shows the list of files being copied:

```
capacity-scheduler.xml
configuration.xml
container-executor.cfg
core-site.xml
hadoop-env.cmd
hadoop-env.sh
hadoop-metrics2.properties
hadoop-policy.xml
hadoop-user-functions.sh.example
hdfs-rbf-site.xml
hdfs-site.xml
https-env.sh
https-log4j.properties
https-site.xml
kms-acls.xml
kms-env.sh
kms-log4j.properties
kms-site.xml
log4j.properties
mapred-env.cmd
mapred-env.sh
mapred-queues.xml.template
mapred-site.xml
example.sh
ssl-client.xml.example
ssl-server.xml.example
user_ec_policies.xml.template
workers
yarn-env.cmd
yarn-env.sh
yarnservice-log4j.properties
yarn-site.xml
```

Después, en el maestro se formatea el NameNode

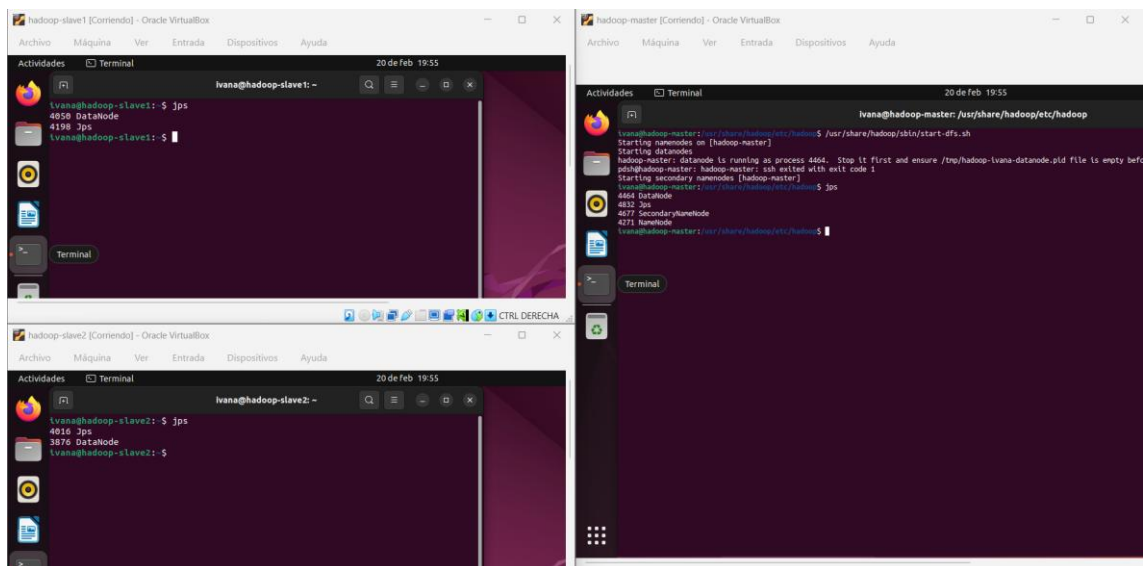
/usr/share/hadoop/bin/hdfs namenode -format.



```
ivana@hadoop-master: /usr/share/hadoop/etc/hadoop
ivana@hadoop-master: /usr/share/hadoop/etc/hadoop$ /usr/share/hadoop/bin/hdfs namenode -format
2026-02-20 19:51:40,777 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = hadoop-master/192.168.1.30
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.3.4
STARTUP_MSG: classpath = /usr/share/hadoop/etc/hadoop:/usr/share/hadoop/share/hadoop/common/lib/kerb-core-1.0.1.jar:/usr/share/hado
jute-3.5.6.jar:/usr/share/hadoop/share/hadoop/common/lib/commons-daemon-1.0.13.jar:/usr/share/hadoop/share/hadoop/common/lib/j2objc-a
share/hadoop/common/lib/token-provider-1.0.1.jar:/usr/share/hadoop/share/hadoop/common/lib/jakarta.activation-api-1.2.1.jar:/usr/sha
```

7. ARRANQUE DE SERVICIOS Y VERIFICACIÓN DE HDFS

- **Arrancar HDFS:** Se ejecuta en el maestro ***/usr/share/hadoop/sbin/start-dfs.sh.***
- **Verificar:** A continuación, se comprueba que los procesos de Hadoop se están ejecutando correctamente en las tres máquinas con el comando Ejecuta ***jps.***



```
hadoop-master [Contenido] - Oracle VirtualBox
ivana@hadoop-master: /usr/share/hadoop/etc/hadoop$ /usr/share/hadoop/sbin/start-dfs.sh
Starting namenodes on [hadoop-master]
Starting datanodes
hadoop-master: datanode is running as process 4464. Stop it first and ensure /tmp/hadoop-ivana-datanode.pid file is empty before
ivana@hadoop-master: /usr/share/hadoop/etc/hadoop$ jps
4464 DataNode
4832 Jps
4877 SecondaryNameNode
4271 NameNode
ivana@hadoop-master: /usr/share/hadoop/etc/hadoop$

hadoop-slave1 [Contenido] - Oracle VirtualBox
ivana@hadoop-slave1: ~$ jps
4058 DataNode
4198 Jps
ivana@hadoop-slave1: ~$

hadoop-slave2 [Contenido] - Oracle VirtualBox
ivana@hadoop-slave2: ~$ jps
4016 Jps
3876 DataNode
ivana@hadoop-slave2: ~$
```

Para verificar HDFS desde la interfaz web, abrimos un navegador ya accedemos a <http://hadoop-master:9870>.

En la pestaña Datanodes debe mostrarse una tabla con los tres nodos del clúster.

Actividades Firefox 20 de feb 20:36

Namenode information x +

No seguro http://hadoop-master:9870/dfshealth.html#tab-overview

Summary

Security is off.
Safemode is off.
18 files and directories, 5 blocks (5 replicated blocks, 0 erasure coded block groups) = 23 total filesystem object(s).
Heap Memory used 91.87 MB of 248 MB Heap Memory. Max Heap Memory is 1.1 GB.
Non Heap Memory used 61.03 MB of 63.38 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	71.83 GB
Configured Remote Capacity:	0 B
DFS Used:	3.25 MB (0%)
Non DFS Used:	50.54 GB
DFS Remaining:	17.56 GB (24.45%)
Block Pool Used:	3.25 MB (0%)
DataNodes usages% (Min/Median/Max/stdDev):	0.00% / 0.01% / 0.01% / 0.00%
Live Nodes	3 (Decommissioned: 0, In Maintenance: 0)
Dead Nodes	0 (Decommissioned: 0, In Maintenance: 0)
Decommissioning Nodes	0
Entering Maintenance Nodes	0

Total Datanode Volume Failures	0 (0 B)
Number of Under-Replicated Blocks	0
Number of Blocks Pending Deletion (including replicas)	0
Block Deletion Start Time	Fri Feb 20 19:54:37 +0100 2026
Last Checkpoint Time	Fri Feb 20 19:51:43 +0100 2026
Enabled Erasure Coding Policies	RS-6-3-1024k

Actividades Firefox 20 de feb 20:39

Namenode information x +

No seguro http://hadoop-master:9870/dfshealth.html#tab-overview

NameNode Journal Status

Current transaction ID: 97

Journal Manager	State
FileJournalManager(root=/mnt/hadoop/tmp/dfs/name)	EditLogFileOutputStream(/mnt/hadoop/tmp/dfs/name/current/edits_inprogress_0000000000000000003)

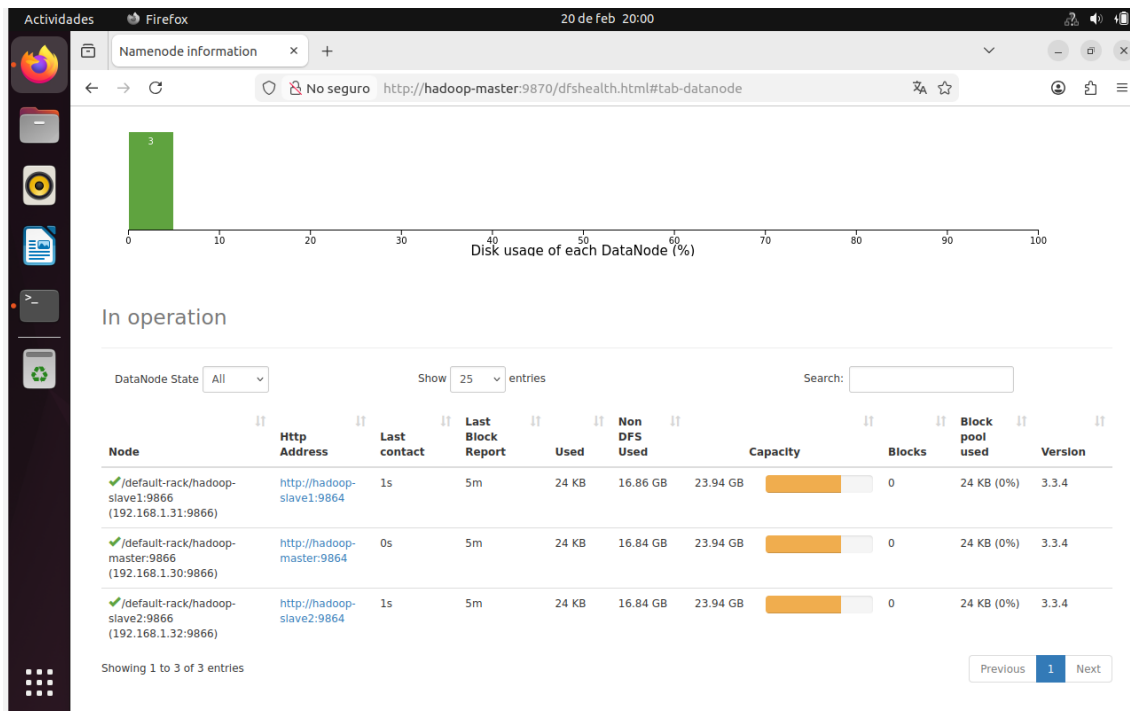
NameNode Storage

Storage Directory	Type	State
/mnt/hadoop/tmp/dfs/name	IMAGE_AND_EDITS	Active

DFS Storage Types

Storage Type	Configured Capacity	Capacity Used	Capacity Remaining	Block Pool Used	Nodes In Service
DISK	71.83 GB	3.25 MB (0%)	17.56 GB (24.45%)	3.25 MB	3

Hadoop, 2022.



The screenshot shows the 'Browse Directories' page in a Firefox browser. The address bar displays 'http://hadoop-master:9870/explorer.html#/user/ivana'. A modal window titled 'File information - workers' is open, showing details for a file named 'workers'.

File information - workers

Download Head the file (first 32K) Tail the file (last 32K)

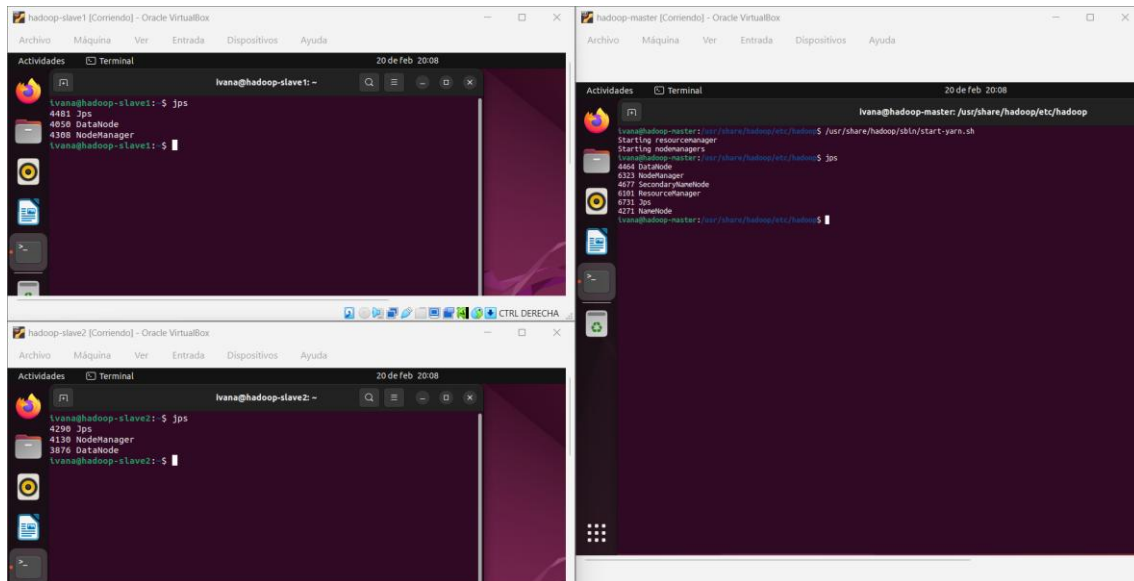
Block information -- Block 0

Block ID: 1073741836
 Block Pool ID: BP-82578553-192.168.1.30-1771613502750
 Generation Stamp: 1012
 Size: 53
 Availability:
 • hadoop-master
 • hadoop-slave1

Close

8. ACTIVACIÓN DEL YARN

Para poder ejecutar programas MapReduce, tendremos que iniciar el gestor de recursos Yarn. En el nodo maestro ejecutamos ***/usr/share/hadoop/sbin/start-yarn.sh*** y después, volvemos a comprobar con ***jps*** que los servicios relacionados con YARN están activos.



9. PRUEBA DE EJECUCIÓN CON WORDCOUNT

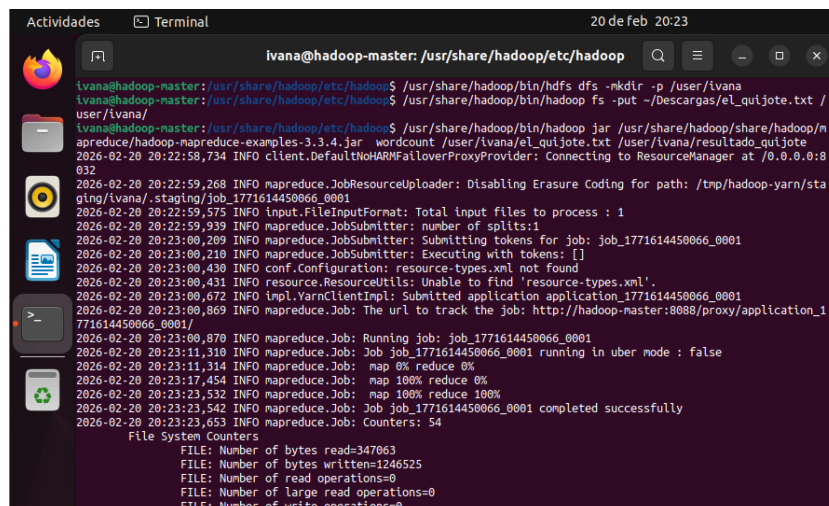
Como prueba final de funcionamiento del clúster, subimos el archivo **el_quijote.txt** al HDFS y ejecutamos el ejemplo de WordCount

Para ello, lo primero que tenemos que hacer es crear la estructura de directorios necesaria dentro de Hadoop con **/usr/share/hadoop/bin/hdfs dfs -mkdir -p /user/ivana**

A continuación, subimos el archivo y lanzamos el trabajo de WordCount:

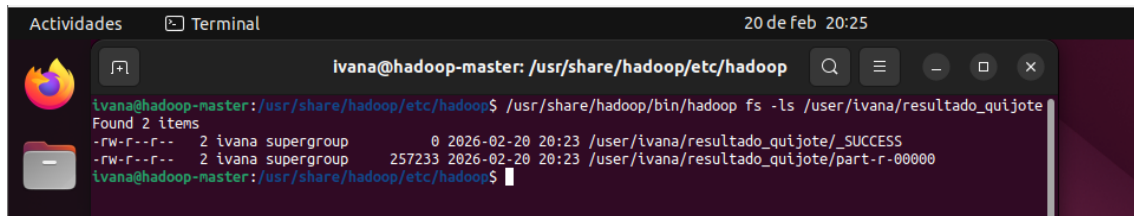
/usr/share/hadoop/bin/hadoop jar /usr/share/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.4.jar wordcount /user/ivana/el_quijote.txt /user/ivana/resultado_quijote

Este proceso pone a trabajar a los nodos esclavos, que se encargan del procesamiento distribuido.



10. VERIFICACIÓN DE SALUD DEL CLÚSTER

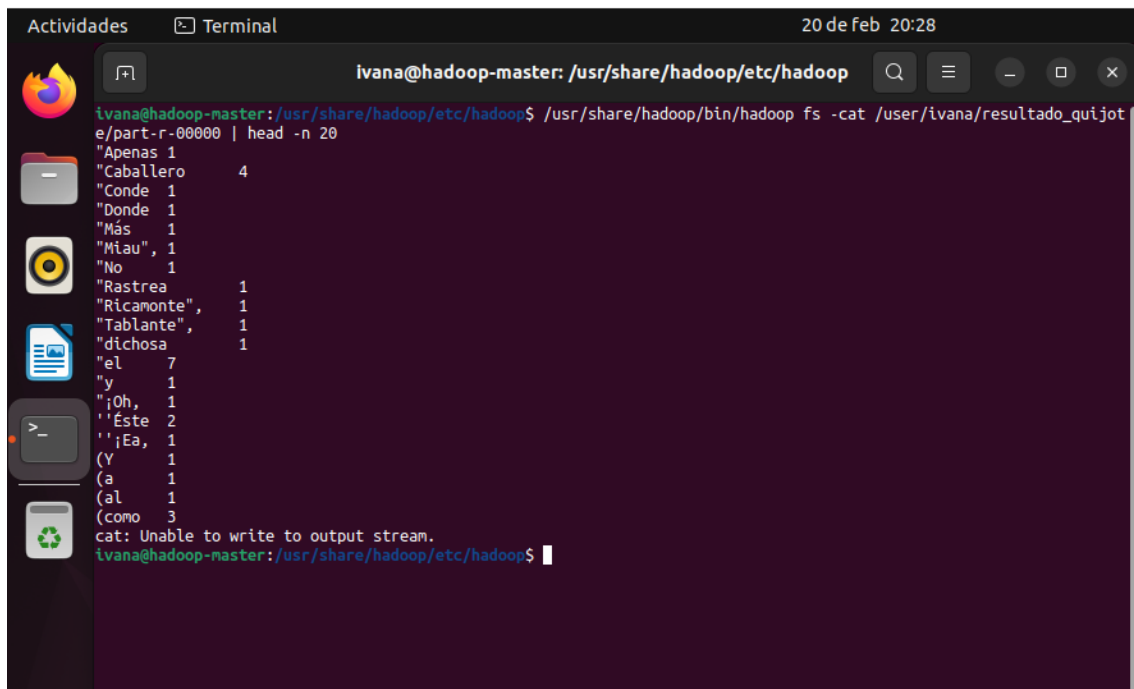
Se comprueba que se han generado los archivos de salida en HDFS y se visualiza parte de su contenido para verificar que WordCount ha funcionado correctamente con los comandos **cat** y **ls**.



```

ivana@hadoop-master: /usr/share/hadoop/etc/hadoop
ivana@hadoop-master: /usr/share/hadoop/etc/hadoop$ /usr/share/hadoop/bin/hadoop fs -ls /user/ivana/resultado_quijote
Found 2 items
-rw-r--r-- 2 ivana supergroup 0 2026-02-20 20:23 /user/ivana/resultado_quijote/_SUCCESS
-rw-r--r-- 2 ivana supergroup 257233 2026-02-20 20:23 /user/ivana/resultado_quijote/part-r-000000
ivana@hadoop-master: /usr/share/hadoop/etc/hadoop$

```



```

ivana@hadoop-master: /usr/share/hadoop/etc/hadoop$ /usr/share/hadoop/bin/hadoop fs -cat /user/ivana/resultado_quijote/part-r-000000 | head -n 20
"Apenas 1
"Caballero 4
"Conde 1
"Donde 1
"Más 1
"Miau", 1
"No 1
"Rastrea 1
"Ricamonte", 1
"Tablante", 1
"dichosa 1
"el 7
"y 1
"¡Oh, 1
"Este 2
"¡Ea, 1
(Y 1
(a 1
(al 1
(como 3
cat: Unable to write to output stream.
ivana@hadoop-master: /usr/share/hadoop/etc/hadoop$

```

1. MONITORIZACIÓN EN EL RESOURCEMANAGER (YARN)

Para revisar el estado de la ejecución en la interfaz web de YARN, accedemos desde el navegador a:

- URL: <http://192.168.1.30:8088>

En esta interfaz debe aparecer la aplicación con estado **FINISHED** y **SUCCEEDED**, mostrando también información de tiempos y recursos utilizados.

Finalmente, para monitorear en tiempo real la ejecución en el ResourceManager, se vuelve a lanzar el WordCount (borrando antes la carpeta de resultados o usando una nueva ruta de salida)

El comando utilizado es el siguiente:

`/usr/share/hadoop/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.4.jar wordcount /user/ivana/el_quijote.txt /user/ivana/resultado_final`

- `http://hadoop-master:8088`

En la pantalla principal aparecerá una tabla con las aplicaciones en ejecución, donde se puede identificar la aplicación «wordcount». En dicha tabla se muestran columnas como **State**, que debería pasar de RUNNING a FINISHED, y **FinalStatus**, que debería indicar SUCCEEDED cuando la ejecución termine correctamente.

The screenshot shows the Hadoop ResourceManager web interface. The 'All Applications' page is active, displaying a table of applications. The application 'wordcount' is listed with the following details:

ID	User	Name	Application Type	Queue	Application Priority	Start Time	Launch Time	Finish Time	State	Final Status	Running Containers	Allocate CPU Vcores
application_1771614450066_0001	ivana	word count	MAPREDUCE	default	0	Fri Feb 20 20:23:00 +0100 2026	Fri Feb 20 20:23:02 +0100 2026	Fri Feb 20 20:23:22 +0100 2026	FINISHED	SUCCEEDED	N/A	N/A

The screenshot shows a detailed view of the 'All Applications' page. The application 'wordcount' is listed with the following details:

startTime	LaunchTime	FinishTime	State	FinalStatus	Running Containers	Allocated CPU Vcores	Allocated Memory MB	Allocated GPUs	Reserved CPU Vcores	Reserved Memory MB	Reserved GPUs	% of Queue	% of Cluster	Progress	Tracking UI	Blacklisted Nodes
Fri Feb 20 20:23:00 +0100 2026	Fri Feb 20 20:23:02 +0100 2026	Fri Feb 20 20:23:22 +0100 2026	FINISHED	SUCCEEDED	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.0	0.0	100%	History	0